

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

1. Apply Backtracking Search for Doctor Shift Assignment

Problem Statement

A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts:

- Morning
- Afternoon
- Night

Given Constraints

1. D1 cannot work Night shift.
2. D2 must work before D3 (Morning < Afternoon < Night).
3. Only one doctor can be assigned to each shift.
4. D3 cannot work Morning shift.
5. All doctors must be assigned exactly one shift.

Step 1: Define Variables and Domains

Each doctor is considered as a variable.

Doctor	Available Shifts
D1	Morning, Afternoon
D2	Morning, Afternoon, Night
D3	Afternoon, Night

Step 2: Backtracking Search Process

Initially, no doctor is assigned.

Initial State

Assignment = { }

Assign D1

Try:

D1 = Morning

Constraint checking:

- D1 cannot work Night → Satisfied
- Shift is available → Satisfied

Current Assignment:

D1 → Morning

Assign D2

Available shifts:

- Afternoon
- Night

Try:

D2 = Afternoon

Constraint checking:

- No two doctors have the same shift → Satisfied
- D2 must work before D3 → Possible

Current Assignment:

D1 → Morning

D2 → Afternoon

Assign D3

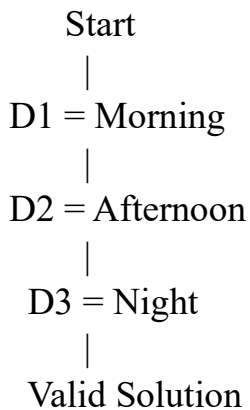
Remaining available shift:

D3 = Night

Constraint checking:

- D3 cannot work Morning → Satisfied
- D2 before D3 → Afternoon < Night → Satisfied
- All doctors assigned exactly one shift → Satisfied

Backtracking Search Tree



Final Valid Schedule

Shift	Assigned Doctor
Morning	D1
Afternoon	D2
Night	D3

2. BFS Search Algorithm for Robot Grid Navigation

Problem Statement

A robot operates in a 5×5 grid environment. It must move from Start (S) to Goal (G) while avoiding obstacles.

Constraints

- Movement allowed: Up, Down, Left, Right
- Cost of each movement = 1
- Cannot move through obstacles
- Manhattan Distance heuristic is given

Grid Representation

Example grid:

```
S . . X .  
. X . X .  
. X . . .  
. . X X .  
. . . G .
```

Where:

- S = Starting position
- G = Goal position
- X = Obstacle
- . = Empty cell

Breadth First Search (BFS) Steps

BFS explores nodes level by level using a queue.

Initial State

Queue:

[(0,0)]

Visited Node:

(0,0)

Step 1: Expand Starting Node (0,0)

Possible movements:

- Down $\rightarrow$ (1,0)
- Right $\rightarrow$ (0,1)

Queue becomes:

[(1,0),(0,1)]

Step 2: Expand (1,0)

New available node:

- (2,0)

Queue:

[(0,1),(2,0)]

Step 3: Expand (0,1)

New available node:

- (0,2)

Queue:

[(2,0),(0,2)]

Step 4: Continue Expansion

The BFS search continues by expanding the nearest unvisited nodes.

Expansion order:

(0,0)

(1,0)

(0,1)

(2,0)

(0,2)

(3,0)

(1,2)

(3,1)

(2,2)

(4,1)

(2,3)

(4,2)

(4,3)

Optimal Path Found

The shortest path from Start to Goal is:

(0,0)

↓

(1,0)

↓

(2,0)

↓

(3,0)

↓

(3,1)



(4,1)



(4,2)



(4,3)

Cost Calculation

Each movement cost = 1

Number of movements = 7

Therefore:

Total Path Cost = 7

3. Online Search for Autonomous Rescue Robot

Problem Statement

An autonomous rescue robot is deployed in a disaster-affected building to find survivors. The robot works in a partially observable environment where it has limited information.

The robot must move from Start (S) to Goal (G) while avoiding obstacles and minimizing cost.

Constraints

- Robot can move Up, Down, Left, and Right.
- Obstacles cannot be crossed.
- Robot can observe only adjacent cells.
- Normal movement cost = 1.
- Risk zones have additional cost = 2.
- Robot must update its knowledge dynamically.

Online Search Process

Step 1: Initial Observation

The robot starts at position S.

It can only observe nearby cells.

The robot creates an initial map using available information.

Possible actions are evaluated.

Step 2: Movement and Observation

The robot moves to the safest available cell.

Movement cost:

Cost = 1

After moving, it observes new surroundings.

If an obstacle is discovered, the robot updates its map.

Step 3: Handling Unknown Areas

The robot discovers unknown cells while moving.

It updates its knowledge:

- Free path → Continue movement
- Obstacle → Avoid path
- Risk zone → Add extra cost and evaluate alternative path

Step 4: Path Replanning

When new information is obtained, the robot performs replanning.

Example:

- Previously planned path becomes blocked.
- Robot searches for another safe path.

Step 5: Reaching the Goal

The robot reaches the survivor location while minimizing total movement cost.

Knowledge Update Table

Step	Observation	Action
1	Detect nearby cells	Select possible path
2	Find obstacle	Update map and avoid
3	Detect risky zone	Calculate additional cost
4	Find new route	Re-plan movement
5	Reach survivor	Complete rescue

Conclusion

The rescue robot uses an online search strategy because the environment is unknown and changes dynamically. It continuously observes surroundings, updates its knowledge, avoids obstacles, considers risk costs, and selects the safest minimum-cost path to reach the survivor.

